TEXT MINING PROJECT : WINE REVIEWS

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ISM6359 Data Mining

TEXT ANALYSIS OF WINE REVIEWS VIA TWITTER

This analysis of wine reviews is a text mining type of performance with initial goals defined such as dependency of price of wine on the review of the taster, best wine producing province, mostly used blend of grapes for preparing wines.

Wine Reviews

130k wine reviews with variety, location, winery, price, and description

AGENDA

- DATA SET
- GOALS OF ANALYSIS
- DATA CLEANING
- CORRELATION ANALYSIS
- ASSOCIATION RULE MINING
- CLUSTERING ANALYSIS (K-Means)
- 3WS
- CONCLUSION

DATA SET

This dataset, in my opinion, has a lot of potential for sentiment analysis and other text-related prediction models

My general goal is to develop a model that can recognize a wine's variety, winery, and location from a description.

The dataset is a corpus of reviews given by consumers via Twitter.

I've collected the data from Kaggle website.
In total, this dataset has 9000 records.

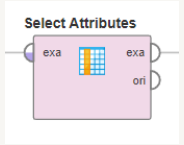
https://www.kaggle.com/datasets/zynicide/wine-reviews

In total, this data consists of 14 attributes such as Wine Origin, Description, Twitter handle of customer who reviewed it. Numerical attributes of this dataset are points rated by customer on the scale of 100 and price of the wine. So, this is an optimal dataset for performing correlation analysis along with the association rule mining and clustering analysis.

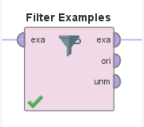
DATA CLEANING

As we got the raw form of data from the tweets of tasters, we need to perform data cleansing. I've made use of Rapid miner operators such as

SELECT ATTRIBUTES



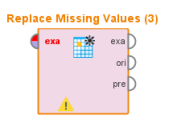
This Operator provides various options for selecting the attributes for the continuation of process. It has options of single, subset, regular expression etc., In this project, we're focusing on single attribute description' There are parameters to invert the selection and include special attributes.



FILTER EXAMPLES

Using this operator, we can filter the attributes that are selected based on criteria defined the user. In this assignment of performing correlation analysis, I've included this operator to filter out any missing values. So, we can define criteria at attribute level.

REPLACING MISSING VALUES

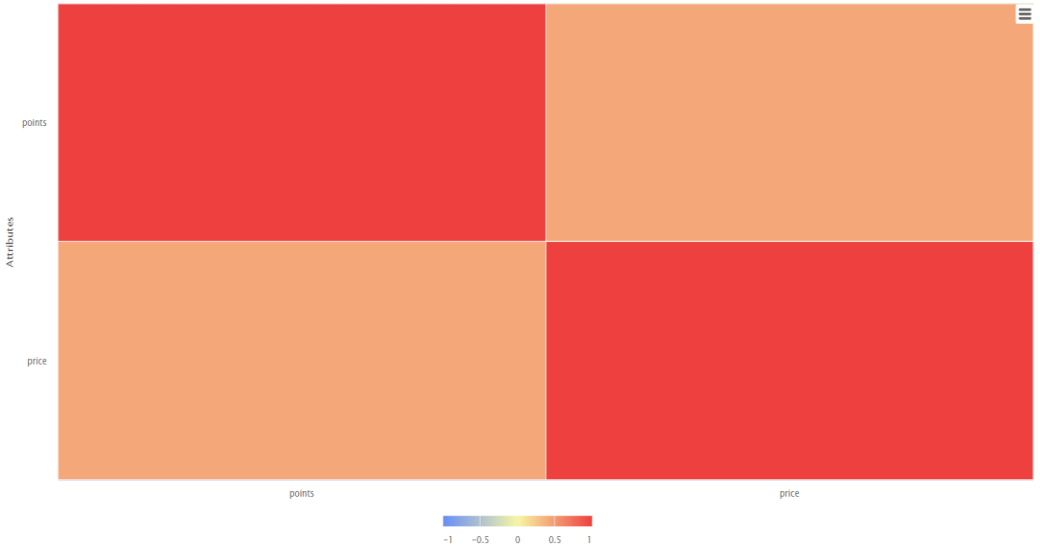


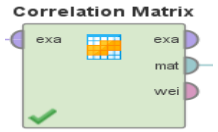
While on the design screen in the operators I searched for "Replace Missing values" and dragged it to the design part where the data set was. After connecting the data set's ‘Exa' to the missing values 'res' I have selected the columns where the data was missing in the attribute filter type to single and replaced the missing values with none, then I connected the replace missing values to the result and run the design which gave me the result where there were "?' are removed.

CORRELATION ANALYSIS

This Correlation Analysis is a statistical method which is used to understand and evaluate the strength of relationship between the significant variables. This has two types of correlation they are High Correlation and Weak Correlation. Here, high correlation refers to multiple variables having very similar relationship between them. Whereas weak correlation refers to multiple variables are hardly related.

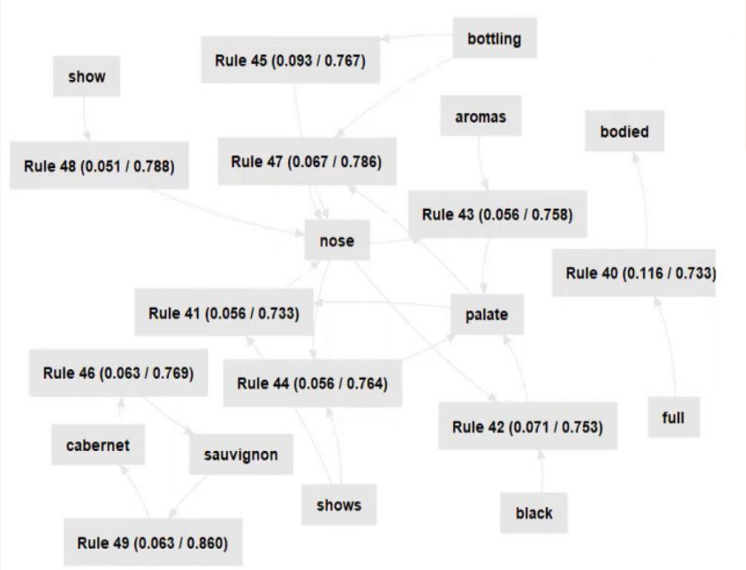
Attributes	points	price
points	1	0.430
price	0.430	1





So, From the above results we can conclude that price and points are weakly correlated.

ASSOCIATION RULE MINING



We use this association analysis to identify interesting associations in multiple large data sets. In this we have two types of relationships that are interesting they are repetitive sets or association rules. Here repetitive sets are the collection of sets that occur frequently and other way to view the interesting relationships are association rules.

AssociationRules

```
Association Rules
[sauvignon] --> [cabernet] (confidence: 0.646)
[white] --> [blend] (confidence: 0.732)
[blanc] --> [sauvignon] (confidence: 0.784)
[pinot] --> [noir] (confidence: 0.815)
[style] --> [bordeaux] (confidence: 0.824)
[style] --> [blend, bordeaux] (confidence: 0.824)
[blend, style] --> [bordeaux] (confidence: 0.824)
[cabernet] --> [sauvignon] (confidence: 0.856)
[sparkling] --> [blend] (confidence: 0.955)
[noir] --> [pinot] (confidence: 0.997)
[style] --> [blend] (confidence: 1.000)
[bordeaux] --> [blend] (confidence: 1.000)
[champagne] --> [blend] (confidence: 1.000)
[franc] --> [cabernet] (confidence: 1.000)
[bordeaux] --> [style] (confidence: 1.000)
[bordeaux] --> [blend, style] (confidence: 1.000)
[blend, bordeaux] --> [style] (confidence: 1.000)
[style, bordeaux] --> [blend] (confidence: 1.000)
[style, white] --> [blend] (confidence: 1.000)
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So, From the above results of association rule mining, we Can state that Bordeaux Blend are the mostly used blend of grapes of wine production.

CLUSTERING ANALYSIS

It is an Unsupervised Machine Learning Technique which is used for detecting or creating clusters of data points in each data set. As different data sets are eliminated, common data sets are grouped into a single cluster. It means that a single cluster has only data points of its kind.

Attribute	cluster_0	cluster_1	cluster_2	cluster_3 ↓
apple	0.001	0.010	0.001	0.040
lemon	0.000	0.002	0.000	0.036
pear	0.000	0.015	0	0.030
lime	0.000	0.004	0	0.026
citrus	0.003	0.014	0	0.026
white	0.003	0.006	0.001	0.026
acidity	0.009	0.006	0.005	0.025
peach	0.000	0.017	0.001	0.024
crisp	0.002	0	0.001	0.022
palate	0.018	0.020	0.018	0.020
fresh	0.008	0.010	0.007	0.019
nose	0.019	0.001	0.007	0.019
chardonnay	0.000	0.002	0	0.019
orange	0.003	0.011	0.004	0.018

PerformanceVector

```
PerformanceVector:
Avg. within centroid distance: -1.009
Avg. within centroid distance_cluster_0: -0.998
Avg. within centroid distance_cluster_1: -1.021
Avg. within centroid distance_cluster_2: -1.008
Avg. within centroid distance_cluster_3: -1.006
Avg. within centroid distance_cluster_4: -1.01
Davies Bouldin: -12.444
```

Cluster Model

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Cluster 0: 611 items
Cluster 1: 1675 items
Cluster 2: 2465 items
Cluster 3: 1848 items
Cluster 4: 2400 items
Total number of items: 8999
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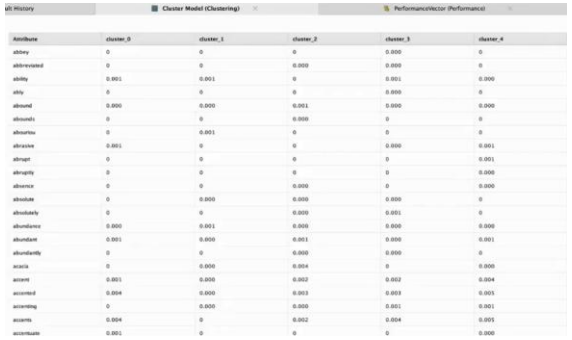
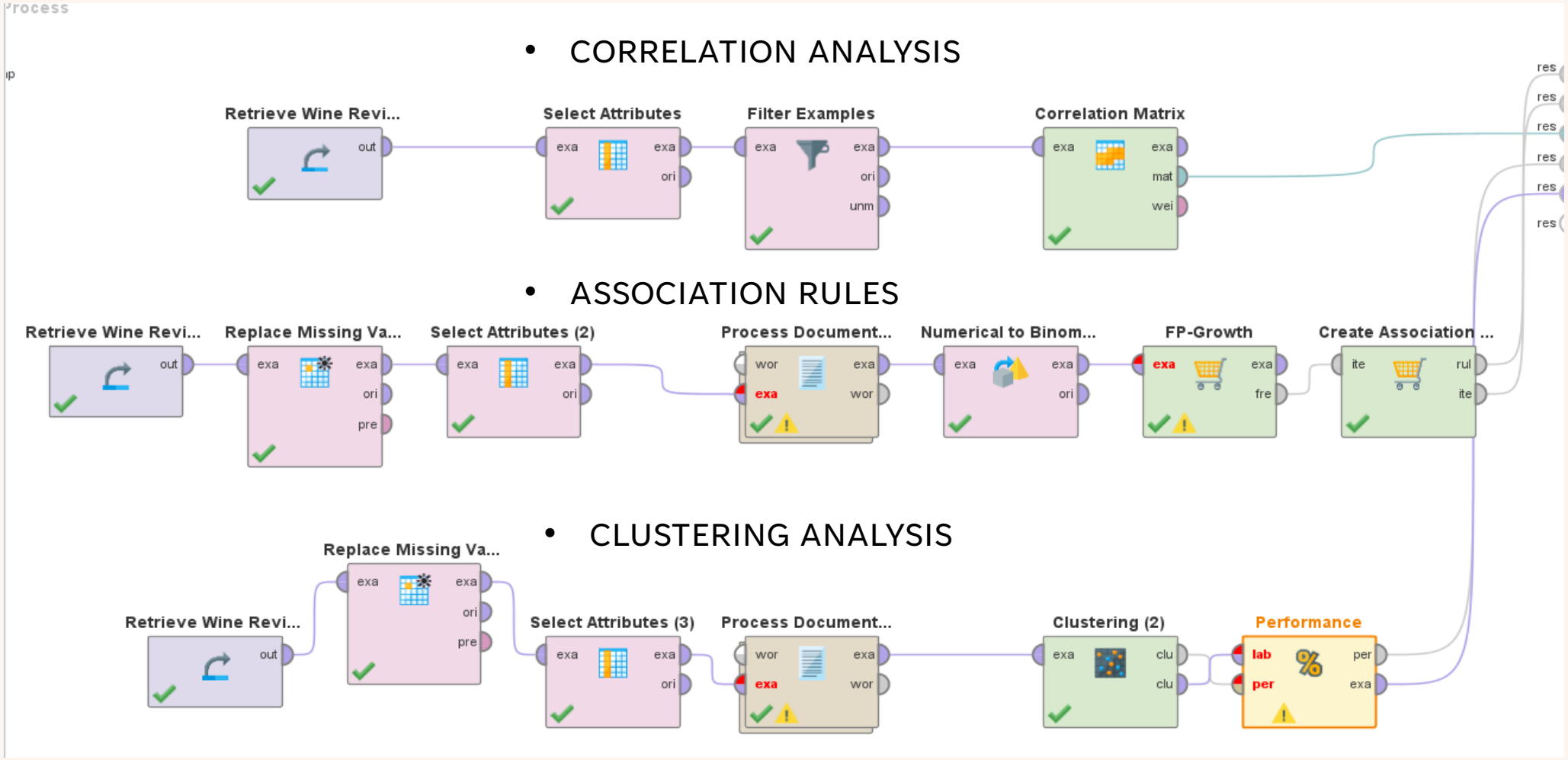


It resulted in 5 clusters, each representing types of wines such as Citrus, fruity, berries, spice and brands cluster.

FINAL PROCESS



3W'S

WHAT WENT WELL

- Successfully implemented the Data Mining Techniques.
- Used Rapid Miner for comprehensible analysis.

WHAT *DID NOT* GO WELL

- I was anticipating more favorable outcomes.

WHAT WOULD I DO DIFFERENTLY?

- Exploring deeper into wine insights.
- Gather a more diversified and cleaner dataset.

CONCLUSION

This data can help someone who wants to establish a winery in a variety of ways. It provides information on how to use funds and vines in different places, as well as how to start a winery with less money.

- Based on the points given by tasters, they can opt to produce a superior tasting wine, and if sales are excellent, they can set the price and distribute it across different locations

- Because there are several steps in the wine-making process, the rules of the association provide information on the most used grapes and what is included in most of the wine-making process

- Using the process description, they can determine which grape variety is most used, as well as limit the equipment required.

- Taste is linked to the process; therefore, a good wine is the result of a good process, which is the result of good procedures in the process. All this information is beneficial to wine startup companies.

EXPERIENCE FROM THIS ANALYSIS

There are many operators through which I have analyzed the data and came to know how a machine learning tool can be used to analyze data

THANK YOU

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